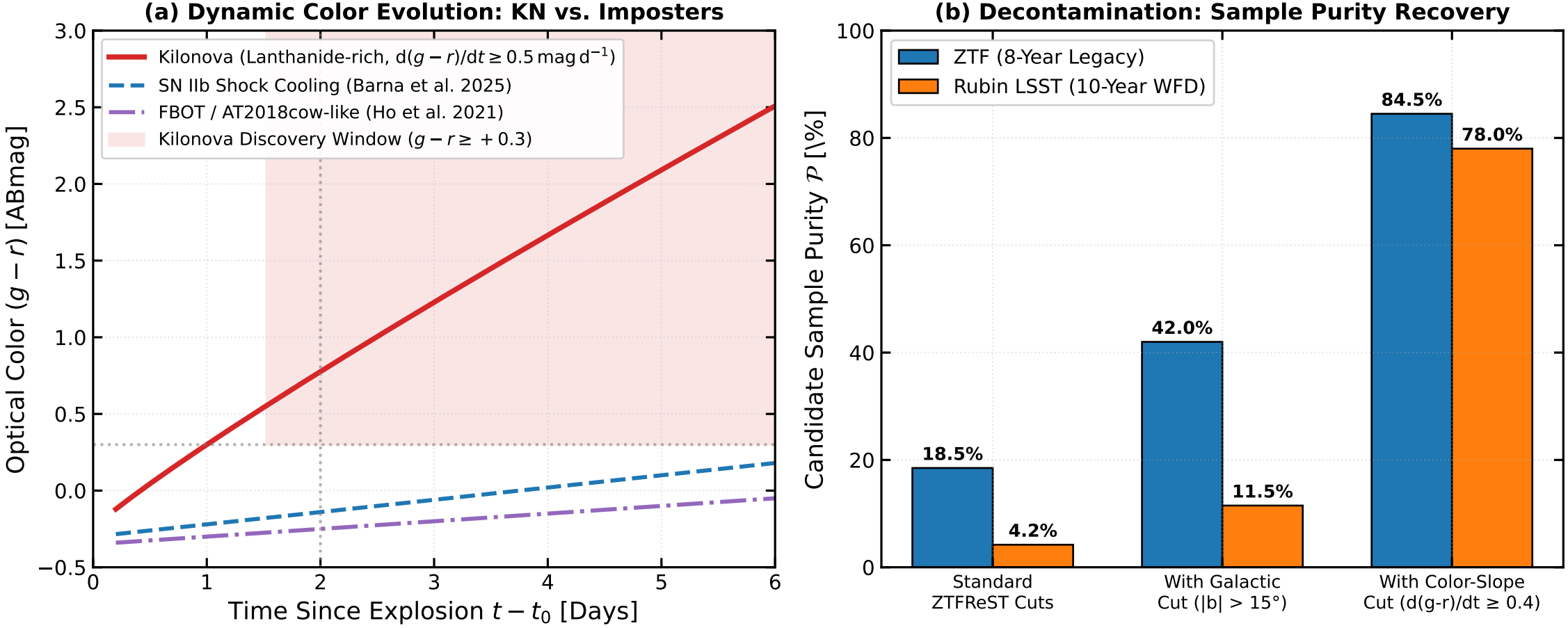
(a) Dynamic Color Evolution: KN vs. Imposters
Kilonova (Lanthanide-rich, d(g − r)/dt ≥ 0.5 mag d⁻¹)
SN IIb Shock Cooling (Barna et al. 2025)
FBOT / AT2018cow-like (Ho et al. 2021)
Kilonova Discovery Window (g − r ≥ +0.3)
Optical Color (g − r) [ABmag]
Time Since Explosion t − t0 [Days]
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
5
6
(b) Decontamination: Sample Purity Recovery
ZTF (8-Year Legacy)
Rubin LSST (10-Year WFD)
Candidate Sample Purity P [%]
100
80
60
40
20
0
18.5%
4.2%
42.0%
11.5%
84.5%
78.0%
Standard
ZTFReST Cuts
With Galactic
Cut (|b| > 15°)
With Color-Slope
Cut (d(g-r)/dt ≥ 0.4)